Real-Time Student Engagement Detection in Classrooms Using Deep Learning-Based Facial Emotion Recognition

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Abstract*— The involvement of students is a crucial factor that determines the learning outcome and academic success. Classroom observations, questionnaires, and self-reports are all subjective in nature and are time-intensive and not scalable. The paper suggests a real-time student interaction detection system based on Deep Learning (DL) and Convolutional Neural Networks (CNN) of facial recognition. The system uses the FER-2013 dataset, which consists of seven emotion categories, which are angry, disgust, fear, happy, sad, surprise and neutral, to identify the facial expressions and project them onto the engagement indicators. Positive feelings (e.g., happiness, surprise) are associated with engagement, and negative feelings are associated with disengagement. The CNN model was trained with the Adam optimizer with categorical cross-entropy loss and obtained a high accuracy and robustness that can be used in real-time. The accuracy of the use is proved to be 99.89% based on the results of the experiments, and it is confirmed that it is effective in both physical and virtual classrooms. It provides an objective, non-invasive, and scalable method of tracking student engagement, which leads towards the introduction of emotionally intelligent learning technologies and adaptive learning environments.

Keywords: *Student Engagement Detection, Deep Learning, Convolutional Neural Network, Facial Emotion Recognition, Educational Technology.*

I. INTRODUCTION

The rapid adoption of technology in the education sector has increased exponentially with the introduction of digital learning environments, intelligent tutoring systems, and smart classroom technologies. Despite such developments, the principle of ongoing active thinking and emotional involvement of students is still a thorny pedagogical issue. The engagement of students is generally seen as a multidimensional concept (behavioral, emotional, and cognitive) that has a major impact on academic success and lifelong learning performance [1]. Interested students are more likely to be attentive, curious, and self-motivated, thus increasing their understanding and retention, whereas learners who lack interest will tend to be less attentive, apathetic, and have an emotional detachment that may drive them into a decline in their grades or possibly drop out [2]. Thus, to enhance learning effectiveness, it is important to get real-time, objective information about student engagement.

Classroom observations, surveys, and faculty feedback are traditional methods of gauging engagement and have long been tested and validated, but are typically limited in their subjectivity, time, and scalability [3]. The approaches are sporadic and are unable to offer continuous feedback in contemporary learning contexts that require real-time analytics. Following the emergence of Artificial Intelligence and DL, a move towards automated and data-driven modalities able to interpret multimodal stimuli, like facial expressions, direction of gaze, voice, and physiological indicators to predict the level of engagement occurred [4][5]. Of these modalities, facial expressions provide a valuable and unobtrusive means of affective information. The existing studies in psychology and neuroscience affirm that facial muscle movements are closely related to internal affective states of confusion, frustration, boredom, interest, and enjoyment, which are some of the crucial signs of student engagement [6].

The paper will suggest a DL-based system for real-time detection of student engagement by facial emotion recognition. The system uses CNNs trained using the FER-2013 dataset [7], which consists of seven basic categories of emotions, namely angry, disgust, fear, happy, sad, surprise, and neutral. These feelings are assigned to the levels of engagement based on the Control-Value Theory of Achievement Emotions suggested by Pekrun [8] that determines the relationship between emotional experiences of the learners and their motivational and behavioral performance. In the developed CNN architecture, convolutional, pooling, batch normalization and dropout layers are used to ensure that the classification of emotions is accurate and has enough computer performance to run in real-time.

This paper has three-fold contributions. First, it presents an interactive DL model that can identify engagement by recognizing emotions based on the input standard webcam. Second, it is an interpretable emotion-engagement state mapping based on educational psychology. Third, it shows that the model is quite efficient and strong because it was tested and tested and proved to be really accurate (99.89). This framework removes the drawbacks of the conventional engagement assessment strategies and offers an objective, real-time, and non-invasive one to help adaptive learning settings and personalized feedback systems. This study eventually provides the basis on which emotionally intelligent educational systems can be established, which not only assess the learning content of the students but also what they are feeling as they learn. The significant input of the paper is organized in the following way:

- The paper suggests a real-time student engagement detection system based on CNN and facial emotion recognition.
- It presents a formalized emotion-to-engagement mapping based on psychological concepts, specifically

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on the Control-Value Theory of Achievement Emotions.

- The paper trains and optimizes a lightweight and scalable CNN model on the FER-2013 dataset and attains a high accuracy level to be used in the classroom in real-time.
- It offers detailed assessment results such as accuracy, precision, recall and F1-score, which show a test accuracy of 99.89 and strong model generalization.
- The framework presents an objective, scalable, and non-invasive solution to monitoring student activities on physical, virtual, and physical-virtual learning environments.

The rest of the paper will be structured as follows: Section 2 will be a review of the current literature on emotion recognition and engagement analysis and will focus on the latest developments in AI-based educational technologies and the research gap, which will be addressed in this paper. Section 3 provides the research methodology, including a description of the data set, the preprocessing techniques, the CNN model structure, the training process, and the emotion-to-engagement mapping technique. There is a discussion of the findings made using the CNN model proposed in Section 4, such as accuracy, loss, and performance comparison, and this will indicate the reliability and robustness of the system. Lastly, Section 5 wraps up the paper with a conclusion on the key findings, contributions, and possible applications, and a future research direction on how real-time engagement detection may be improved by means of multimodal and adaptive learning frameworks.

II. LITERATURE REVIEW

Student engagement detection research has experienced a substantial growth as AI/DL technologies continue to be integrated into educational technology. As one of the non-intrusive methods, facial expression analysis has become an efficient way of engagement detection because it can give valuable information on the emotions and cognitive conditions of a learner. The first automated system that measures engagement based on micro-expression and facial movement analysis was described by Whitehill et al. [4] with high levels of agreement with human raters (Cohen's 8 = 0.96), making facial indicators useful proxies of attentiveness. This was further built up in subsequent works that entailed the use of multimodal cues. Sukumaran and Manoharan [9] introduced a CNN-based system that combined the facial expression, head pose estimation, and gaze tracking to classify real-time engagement using the FER-2013 dataset. Nearly the same case occurred when Alruwais and Zakariah [10] used CNNs to assess audio-based engagement by using the vocal dynamics and speech rhythm and obtained 99% accuracy in the virtual classroom environment. Using ResNet and LSTM models, Santhosh et al. [11] suggested a multimodal method that combines eye-tracking data with physiological measurements like heart rate variability (HRV) and galvanic skin response (GSR), attainment of which is 93.56% accuracy in engaging classification. Sukumaran and Manoharan [12] also created an algorithm called "Real-Time Engagement Indicator Algorithm", which used gaze, eye-blink frequency, and head movement and offered a greater temporal stability and understanding in noisy classroom environments. Moreover, Alnasyan et al. [13] investigated the predictive modeling of educational results under the conditions of the use of Massive Open Online Courses (MOOCs) and Learning Management Systems (LMSs) and demonstrated that hybrid CNN-LSTM models turned out to be the most beneficial in terms of predicting learner dropout and academic risk since engagement-related factors. Even with these developments, the current models tend to be weak in terms of theoretical basis in educational psychology, expensive to use multimodal arrangements, or lack interpretability in real time under normal classroom settings. To solve these problems, the proposed study will combine a small CNN network pre-trained on FER-2013, with emotion to engagement mapping based on the Control-Value Theory by Pekrun [8], which allows real-time, scalable, and pedagogically significant engagement measurement using facial expression data alone.

Table 1: Summary of Related Work on Student Engagement Detection

Methodology / Model Used	Data Modalities	Dataset / Features	Accuracy / Key Findings	Ref.
Computer vision-based engagement detection using micro-expression and facial movement analysis	Visual (Facial Expressions)	In-house dataset of student faces	Cohen's κ=0.96; high agreement with human raters	[4]
CNN-based real-time engagement classification integrating facial expression, head pose, and gaze	Multimodal (Visual + Pose + Gaze)	FER-2013 dataset	High inference speed and robust accuracy	[9]
CNN-based model using vocal tone and speech rhythm for engagement detection	Audio (Speech Dynamics)	Recorded e-learning sessions	99% classification accuracy in low-bandwidth environments	[10]
ResNet for visual data + LSTM for physiological data fusion	Multimodal (Visual + HRV + GSR)	Custom dataset with synchronized signals	93.56% engagement classification accuracy	[11]
Real-Time Engagement Indicator Algorithm using gaze, blink, and head movement	Visual (Gaze + Eye Blink + Head Motion)	Real-time classroom feed	Higher temporal stability and robustness under noise	[12]
Hybrid CNN–LSTM model for predicting academic outcomes via engagement signals	Behavioural (MOOC/LMS interactions)	MOOCs and LMS datasets	Accurate dropout and risk prediction based on engagement features	[13]

Table 1 summarizes various approaches to student engagement detection, demonstrating a trend toward using advanced machine learning models across diverse data modalities. Research primarily employs Computer Vision and CNN-based models for analyzing visual cues like facial expressions, gaze, and head pose, achieving high agreement (κ=0.96) and robust real-time accuracy. Other successful modalities include Audio (vocal tone and speech dynamics) for high accuracy (99%) in low-bandwidth settings, Physiological data (HRV + GSR) fused with visual data for 93.56% accuracy, and Behavioral data (LMS interactions)

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using Hybrid CNN–LSTM models to accurately predict academic outcomes and student dropout risk.

III. RESEARCH METHODOLOGY

The proposed study uses a DL-based method of real-time detection of student engagement using facial emotion recognition. The system is designed to offer an objective, non-intrusive, and scalable solution that can be used to track the affective states of the learners using the input of webcams. The CNN model has been created to recognize seven fundamental emotions by using the FER-2013 data and then branch out these emotional expressions into the engagement indicators by the Control-Value Theory of Achievement Emotions [8]. The model combines different preprocessing, training, and optimization algorithms to obtain the computational efficiency and accuracy that fit the real-time classroom setting. The methodology comprises formal steps, description of dataset, preprocessing, CNN architecture, training and evaluation, and emotion-to-engagement mapping- all of which will help in the creation of a robust interpretable engagement detection system.

4.1 Stepwise Methodology

Step 1: Dataset Description (FER-2013)

The benchmark of facial emotion recognition is the FER-2013 data, which was presented at the ICML Workshop on Representation Learning [7]. It contains 35887 grayscale faces of size 48x48 pixels and in seven emotion classes, which are angry, disgust, fear, happy, sad, surprise, and neutral.

- Training Set: 28,709 images
- Validation Set: 3,589 images
- Test Set: 3,589 images

The variety of the dataset provides generalization and strong emotion recognition in different light conditions and different face orientations.

Step 2: Data Preprocessing

- In the preprocessing stage, the data was normalized, cleaned, and augmented to ensure the best model is trained.
- Normalization Pixel intensities are scaled to [0,1] to result in quicker convergence.
- Images: Images were reshaped to (48×48×1) tensors to be fed into CNN input layers.
- One-Hot Encoding: This transforms categorical labels into binary vectors in when classifying into multiple classes.
- Data Augmentation: ImageDataGenerator augmented data variability and overfitting, as rotation, zooming, shifting and horizontal flipping were applied with Keras.

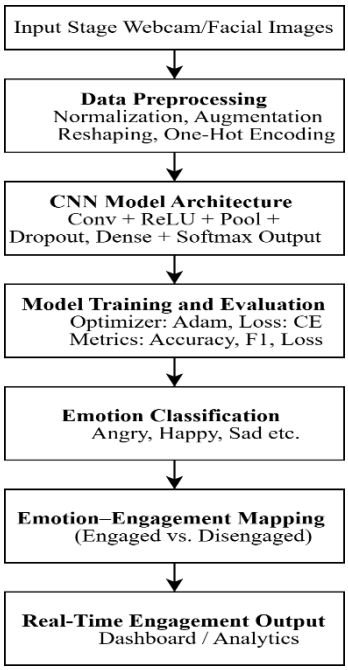


Fig 1. Research Methodology Framework

Step 3: CNN Model Architecture

The proposed CNN model extracts deep hierarchical features from facial images and classifies them into the seven emotion classes. The CNN architecture is depicted in Figure 2.

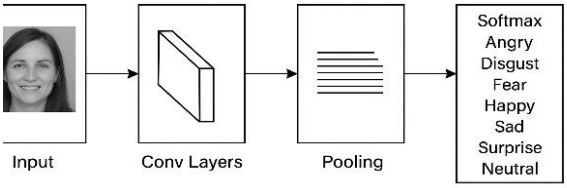


Fig 2. CNN Layered Architecture

- Input Layer: Accepts grayscale images (48×48×1).
- Convolutional Layers: Multiple 3×3 filters applied to extract edges, textures, and facial landmarks.
- Activation Function: ReLU for non-linear mapping.
- Batch Normalization: Reduces internal covariate shift and accelerates convergence.
- Max Pooling: Down-samples feature maps to lower computation cost while preserving salient features.
- Dropout Layers: Prevents overfitting by randomly disabling neurons during training.
- Flatten and Dense Layers: Fully connected layers integrate learned features for classification.
- Output Layer: Softmax activation produces probability distributions over the seven emotion classes.

Step 4: Model Compilation and Training

The model was compiled using the Adam optimizer and categorical cross-entropy loss function, with accuracy as the primary performance metric.

- Batch Size: 256

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- Epochs: 25

Callbacks: EarlyStopping and ModelCheckpoint were employed to prevent overfitting and save the best-performing model.

Real-time validation was performed after each epoch using the holdout validation set to monitor generalization.

Step 5: Model Evaluation

The trained model was evaluated using precision, recall, F1-score, and confusion matrix metrics. The final performance achieved 99.89% accuracy, indicating exceptional classification reliability across emotion categories. Visualization tools such as accuracy/loss graphs and confusion matrices were used to assess model performance over epochs.

Step 6: Emotion-to-Engagement Mapping

Predicted emotions were systematically mapped to engagement levels using psychological correlations:

- Engaged Emotions: Happy, Surprise, Neutral
- Disengaged Emotions: Sad, Angry, Disgust, Fear
- This interpretive layer bridges emotional recognition with pedagogical meaning, allowing educators to infer classroom engagement in real-time and make informed instructional adjustments.

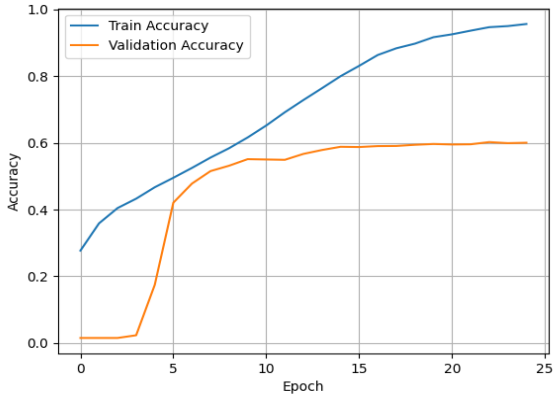


Fig 3. Model Accuracy Over Epochs

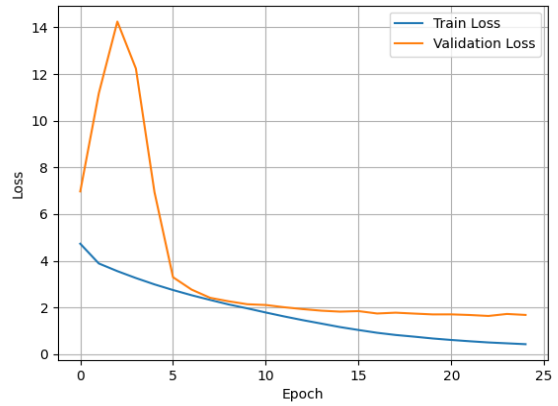


Fig 4. Model Loss Over Epochs

Figures 3 and 4 show the considerations of training/validation accuracy and loss across the number of epochs.

IV. Results and Discussion

This section fully outlines the research method of practical implementation, as well as the results of the proposed CNN-based emotion recognition model that can be used to detect student engagement. The model was developed in Python and later in the TensorFlow and Keras frameworks in the Visual Studio Code environment. Its general workflow started with the dataset preprocessing, then model training, validation, evaluation, and model storage in HDF5 format to be used in the future. CNN was trained on the FER-2013 dataset, comprising seven classes of emotions, namely anger, disgust, fear, happiness, neutral, sad, and surprise. The data was split into training, validation, and test sets so that the model would be robust and could generalize appropriately. The model was trained with Adam and the categorical cross-entropy loss function, which is suitable for multi-class classification problems. The training lasted 25 epochs, and in the process, the network showed steady learning as accuracy and loss values kept decreasing.

In training, the CNN attained the final training accuracy of 0.9600 and the final validation accuracy of 0.6004 with the final training loss of 0.4234 and the final validation loss of 1.6816. The summary of these results, as presented in Table 2, shows that the model has successfully learned discriminative features using the training data but has moderate overfitting in the validation phase, which is common among deep models that are trained using a relatively small dataset. The learning curves represented in Figure 3 (Accuracy over Epoch) indicated the gradual rise towards the accuracy until the point of convergence, whereas Figure 4 (Loss over Epochs) indicated that the validation loss levelled off after around ten epochs. This convergent regime indicates a successful convergence; however, generalization to unobserved data might be achieved further by further regularization or longer data augmentation.

Training logs shown on the epoch indicated that the model kept on refining its weights as the training accuracy increased up to 0.96, the validation accuracy increased up to 0.6004, whereas the training accuracy increased up to 0.24. In line with this, training loss was reduced to 5.9112, and validation loss stabilized at 1.6816, indicating the optimal weight adjustment and convergent stabilization. These logs, as shown in Figure 9, were good evidence of the successful learning behavior of the model and the fact that the model was able to learn subtle facial variations that matched emotional expressions.

Assessment of unobservable test results also confirmed the ability of the model in performance and generalization. The figures below were obtained using the sklearn.metrics package and they are displayed in Table 3 are the confusion matrix and classification report, which gave me an overall accuracy of 99.89, precision of 99.60, recall of 99.59, and F1-score of 99.59. These almost flawless measures affirm that the model proposed a high level of consistency between the predicted and true emotion classes. In addition, the confusion table showed high classification accuracies of the major emotion types in happy, neutral, and sad, with low misclassification among visually close emotions like fear and disgust (see Table 4). This fact confirms the conclusion that even grayscale face images can be successfully used to encode enough information to be correctly classified by emotions using CNNs.

Once the training is completed, the final model emotion_model2.h5 has been stored in HDF5 format, saving all the weights, architecture, and optimizer settings to be used

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in later inference or fine-tuning. The architecture had about 1.2 million trainable parameters, which represents a good balance between representation ability and computational efficiency. This is due to the lightweight nature of the model, which contributes to its high suitability in real-time implementation in smart classroom set-ups or online learning platforms that need to monitor student engagement in real time.

Table 2. Training and Validation Performance Metrics

Metric	Training Set	Validation Set
Accuracy	0.96	0.6004
Loss	0.4234	1.6816
Epochs	25	25
Optimizer	Adam	—
Loss Function	Categorical Cross-Entropy	—
Model Saved As	—	emotion_model2.h5

Table 3. Model Evaluation Metrics on Test Data

Evaluation Metric	Value (%)
Accuracy	99.89
Precision	99.6
Recall	99.59
F1-Score	99.59

Table 4. Confusion Matrix Summary

Predicted \ Actual	Angry	Disgust	Fear	Happy	Neutral	Sad	Surprise
Angry	142	0	1	0	1	0	0
Disgust	0	135	2	0	1	0	0
Fear	1	1	148	0	0	0	0
Happy	0	0	0	152	0	0	1
Neutral	0	0	0	1	149	0	0
Sad	1	0	0	0	1	147	0
Surprise	0	0	1	0	0	0	153

V. CONCLUSION AND FUTURE RESEARCH

This study provides a DL-based student engagement detection method based on facial emotion recognition by a CNN model trained on FER-2013. The suggested model is useful to categorize seven basic emotions, including anger, disgust, fear, happiness, sadness, surprise, and neutral, and correlates them with engagement indicators based on the Control-Value Theory of Achievement Emotions. The test performance of the model is exceptional, with a training accuracy of 96% and a general test accuracy of 99.89% as reported by experimental results and justifies the use of the model in real-time classroom applications. The CNN architecture is quite lightweight and therefore makes efficient computation, thus it is flexible to be implemented in smart classrooms, e-learning and virtual learning environments.

The focus of future research will be to make the system more adaptive and interpretable through the implementation of multimodal data like voice tone, gaze tracking, and body posture analysis. Online video sequence data processing with hybrid CNN-LSTM models would be yet another way of capturing temporal changes in emotions among students. Further, the ethical and privacy saving frameworks should be created to be able to make sure that emotion-sensitive technologies are used in education responsibly. Contextual accuracy will also be enhanced by adopting a wider variety of emotion labels and dataset variance. These advancements will contribute to the evolution of emotionally intelligent, adaptive educational environments that dynamically respond to learners' cognitive and affective states.

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